PRASANNA REDDY PULAKURTHI

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Education

Rochester Institute of Technology — Rochester, NY
Doctor of Philosophy (Ph.D.) – Electrical and Computer Engineering | ***GPA: 3.93/4.0*** — *Aug 2019 to Aug 2025 (expected)*

Rochester Institute of Technology — Rochester, NY
Master of Science (MS) – Electrical Engineering | ***GPA: 4.0/4.0*** — *Aug 2017 to Jul 2019*

PESIT Bangalore South Campus — Bangalore, India
Bachelor of Engineering (BE) – Electronics and Communications Engineering — *Aug 2013 to Jun 2017*

Work Experience

Rochester Institute of Technology
Ph.D. Researcher, ML & Computer Vision — *May 2024 to Aug 2024*

- Developed two novel **data augmentation** techniques (1. Dual-region and 2. Shuffle PatchMix) to improve model robustness and generalization. Demonstrated improvements in performance on domain adaptation datasets such as PACS by up to **+7.3 pp** (percentage points), and person re-identification datasets such as Market-1501 by up to **+6.91 pp**.

DEVCOM Army Research Laboratory (via Booz Allen Hamilton and MBO Partners)
Perception Engineer Intern — *May 2023 to Aug 2023*

- Advanced Generative Models: Developed a GAN variant combining neural-architecture search, PMish activation, bounded MMD repulsive loss, and adaptive rank decomposition, achieving a reduction of FID scores by **34%** (CIFAR-10) and **28%** (STL-10) while compressing the generator **4–10**×, enabling real-time edge deployment.
- Data Generation: Generated new synthetic gesture video data for Human Action Recognition using Generative Adversarial Network and significantly lifted human-action recognition accuracy across ground (**+4.4 pp**) & aerial (**+7.7 pp**) views.

DEVCOM Army Research Laboratory (via Booz Allen Hamilton and MBO Partners)
Perception Engineer Intern — *May 2022 to Aug 2022*

- Developed domain adaptation techniques to improve model robustness across different digit domains, lifting cross-domain accuracy by up to **+2.56 pp** on difficult shifts.
- Globally-optimal Iterative ML Classifier: Formulated a fast, globally-optimal Iterative Maximum Likelihood Classifier by deriving a fixed-point solver from a regularized ML objective, achieving ∼3× faster convergence compared to gradient descent while matching or exceeding accuracy across both linear and non-linear separable data.

Qualcomm Internship – Virtual Reality (VR) Team
Engineering Intern — *May 2021 to Aug 2021*

- Developed a novel temperature-compensation algorithm enhancing the accuracy of the 6DOF 4Tx ultrasound VR controller.
- Implemented the algorithm in C, improving the root mean squared error (RMSE) by **+11.29 pp** on synthetic log data.

Microsoft Internship – Iris Recognition Team for HoloLens 2 Augmented Reality (AR) Team
Engineering Intern — *May 2020 to Aug 2020*

- Optimized Gabor filter frequencies in Iris Recognition pipeline, significantly enhancing signature uniqueness.
- Developed GAN-based Super Resolution Network to enhance iris image quality, maintaining key Gabor frequencies, improving recognition accuracy on lower-quality images.

Oak Ridge National Laboratory
Research Assistant — *May 2018 to Jul 2019*

- Designed and executed an unmanned aerial system (UAS) data collection to capture aerial imagery of a building casting shadows on 14 differently colored panels, both in and out of shadow. [Link]
- Developed six algorithms for shadow detection in aerial images using color transforms and machine learning techniques, leveraging collected data. Improved upon traditional approaches by **+1.51%** in accuracy and **+2.4%** in F-score.

Technical Skills

- **Programming Languages**: C, Python (NumPy, Pandas), HTML/CSS, MATLAB
- **Machine Learning Frameworks/Libraries**: PyTorch, TensorFlow, Keras, Scikit-Learn, OpenCV, LLM, NLP, Transformers
- **Tools**: Docker, DIRSIG, ENVI, Adobe Photoshop, LaTeX, Git, Linux, SQL, Microsoft suite (Excel, Word, PowerPoint)

Publications

- **Prasanna Reddy Pulakurthi**, Jiamian Wang, MAJID RABBANI, Sohail Dianat, Raghuveer Rao, Zhiqiang Tao, "X-CoT: Explainable Text-to-Video Retrieval via Large Language Models (LLM) based Chain-of-Thought Reasoning," *EMNLP* 2025.
- Jiamian Wang, **Prasanna Reddy Pulakurthi**, Pichao WANG, Dongfang Liu, Sohail Dianat, MAJID RABBANI, Raghuveer Rao, Zhiqiang Tao, "Automatic Instruction Programming for Multimodel Large Language Models (MLLM) Based Text-Video Retrieval," *EMNLP* 2025.
- **Prasanna Reddy Pulakurthi**, Majid Rabbani, Jamison Heard, Sohail Dianat, Celso M. de Melo, and Raghuveer Rao, "Shuffle PatchMix Augmentation with Confidence Margin Weighted Pseudo Labels for Enhanced Source-Free Domain Adaptation," *IEEE International Conference on Image Processing (ICIP)*, 2025. [Link]
- **Prasanna Reddy Pulakurthi**, Majid Rabbani, Celso M. de Melo, Sohail A. Dianat, and Raghuveer M. Rao. "Effective dual-region augmentation for reduced reliance on large amounts of labeled data," *Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications III*. SPIE, 2025. [Link]
- **Prasanna Reddy Pulakurthi**, Mahsa Mozaffari, Sohail Dianat, Jamison Heard, Raghuveer Rao, and Majid Rabbani, "Enhancing GANs with MMD Neural Architecture Search, PMish Activation Function and Adaptive Rank Decomposition," *IEEE Access*, 2024, doi: 10.1109/ACCESS.2024.3485557. [Link]
- **Prasanna Reddy Pulakurthi**, Celso M De Melo, Raghuveer Rao, and Majid Rabbani, "Enhancing human action recognition with GAN-based data augmentation," *Synthetic Data for Artificial Intelligence and Machine Learning: Tools, Techniques, and Applications II*. Vol. 13035. SPIE, 2024. [Link]
- **Prasanna Reddy Pulakurthi**, Mahsa Mozaffari, Sohail A. Dianat, Majid Rabbani, Jamison Heard, and Raghuveer Rao, "Enhancing GAN Performance through Neural Architecture Search and Tensor Decomposition," *IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, 2024. [Link]
- **Prasanna Reddy Pulakurthi**, Sohail A. Dianat, Majid Rabbani, Suya You, and Raghuveer M. Rao, "A globally optimal fast iterative linear maximum likelihood classifier," *Electronic Imaging*, 2023, pp 172-1 - 172-5. [Link]
- **Prasanna Reddy Pulakurthi**, Sohail A. Dianat, Majid Rabbani, Suya You, and Raghuveer M. Rao, "Unsupervised domain adaptation using feature aligned maximum classifier discrepancy," Applications of Machine Learning. SPIE, 2022. [Link]
- **Prasanna Reddy Pulakurthi**, "Shadow Detection in Aerial Images using Machine Learning" (2019), Thesis, Rochester Institute of Technology. [Link]

Projects

Real-Time Attendance System using Image Processing [Link]

- Developed a real-time automatic attendance system using OpenCV for face detection, Python-implemented Binarized Statistical Image Features for face recognition, and a Kivy-based app interfacing via sockets and PHP with an Apache-hosted GUI displaying attendance data stored in MySQL.

Automatic License Plate Detection and Recognition [Link]

- Built an automatic number plate detection and recognition system using edge and histogram-based image processing methods for detection and neural networks for recognition, coded in MATLAB.

Machine Learning and AI Projects

- Applied Generative Adversarial Networks (GANs) to unsupervised domain adaptation, image-to-image translation, and human action recognition tasks. [Link]
- Developed Invertible Neural Networks (INNs) to perform super-resolution and image denoising. [Link]
- Designed a full-stack website Vision Captioning Assistant (V-CAT) AI [Link]
- Built a full-stack website for ASL Sign Tutoring (Signmentor AI) [Link]

Image and Signal Processing Projects

- Implemented several image and signal processing algorithms, such as image resizing, noise filtering, image enhancement, image segmentation, camera calibration, stereo reconstruction, affine, and projective transforms.

Coursework

- **Signal & Image Processing**: Digital Signal Processing (EEE678), Digital Image Processing (EEE779), Multi-view Imaging (IMGS712)
- **Machine Learning**: Deep Learning Independent Study (EEE799), AI Explorations (EEE 647), Pattern Recognition (EEE670)
- **Communications**: Information Theory Coding (EEE794), Image and Video Compression (EEE781)
- **Machine Learning Course**: Machine Learning by Stanford University on Coursera